products_02-automata_ORDERING

2 Kalakriti Automata (laser): ordering guide (what to buy, where, in what order)

> Companion to `MANUFACTURING_MANUAL.md`. Main caveat for this project: frame layouts provisional. Prices and stock are from LCSC on the run date; vendors, shipping and customs rules change: verify before paying. General rules and vendor list: `manuals/ORDERING_GUIDE.md`.

1. Order of work

1. Read the manual and the caveat above.
2. No custom circuit board: nothing to order from JLCPCB for this project.
3. While the boards travel, order the other items (sections 3 and 5) and make the printed or laser-cut parts.
4. Assemble, program and test only after everything has arrived; follow the manual.

3. Bought-in and mechanical items

2. Materials and hardware (per kit)

- 3 mm **E1-grade MDF** (formaldehyde class) sheets; measure real thickness (3.0 +-0.2 mm assumed).
- 4 mm dowel (crank shaft) and 6 mm dowel (cam follower), 2 small bearings or smooth holes, rubber band or string, PVA wood glue, water-based paint or lacquer, crank knob, sandpaper.
- Cost about Rs 220 per kit at volume (guess); prototype set about Rs 940.

4. Confidence (read before you order)

- **Order confidence: Medium-High**: no electronic parts to source: local laser-cutting or printing.
- **Works-as-intended confidence: Medium**: Laser cut files regenerated with correct kerf compensation (outlines outward, holes inward, gear backlash only once); cam and gear checks PASS; frame layouts and figure artwork are still first-pass.
- **First thing to do:** Cut one kit in scrap MDF and run the mechanism.
- Rubric and the reasons for every project: `costing/CONFIDENCE.md`. These are judgements, not measurements: nothing here has been built.

5. Where to buy and whom to approach

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
Printed circuit board and SMD assembly	JLCPCB is the practical choice for 5 to 100 boards; local PCB shops quote per project (ask for the Gerber ZIP price)	**JLCPCB** (jlcpcb.com), PCBWay (pcbway.com)
Connectors, headers, passives, ICs by LCSC number	element14 India, Mouser India, DigiKey India (genuine, GST invoice, higher price); local	**LCSC** (lcsc.com) with the same cart as JLCPCB

Item type	India (fast, GST invoice)	Overseas (cheaper, 1 to 3 weeks, customs)
	electronics markets for common parts	
Modules: ESP32 dev boards, SIM7080G, LoRa (RA-08 / RAK3172), pH / EC / DO modules, relay modules, load cells, ultrasonic sensors	Robu.in, Robocraze, ThinkRobotics, Sunrom Electronics, Amazon.in	AliExpress, Seeed Studio, Waveshare
Motors, stepper drivers, wheels	Robu.in, Robocraze, local robotics shops	StepperOnline (omc-stepperonline.com)
Li-ion / LiFePO4 cells and packs, solar panels, MPPT controllers	Authorised battery and solar dealers, Amazon.in (buy protected cells with an invoice)	not recommended by post (shipping rules for batteries)
Laser-cut and 3D-printed parts	Local laser-cutting and 3D-printing shops (search IndiaMART, Justdial or maker spaces): send the SVG/DXF/STL from the repository and ask for a quote per piece	JLC3DP (jlc3dp.com), PCBWay 3D printing
Fasteners, tubing, hardware, enclosures	Local hardware, Robu.in, Amazon.in	n/a

How to approach a vendor: send the project name, the exact file (BOM, CPL, Gerber ZIP, STL or SVG), the quantity, and ask for the price, the lead time, the GST invoice and the warranty or return terms. Keep the reply for your records.